

CS 662 – NLP: Homework #5 Critical Reading of Popular Press

The popular press article that I found was “Amazon Plans to Sell Software that Reads Medical Records”, written by Steven Musil, published on *C Net* on November 27th, 2018.¹ This popular press article was written after the *Wall Street Journal (WSJ)* article “Big Tech Expands Footprint in Health”, written by Melanie Evans and Laura Stevens, published also on November 27th, 2018.² Both of these articles were based off a blog post published by Amazon titled “Introducing Medical Language Processing with Amazon Comprehend Medical”, written by Dr. Taha A. Kass-Hout and Dr. Matt Wood, also published on the same day as the other two articles, but prior to both of them.³ I found it particularly interesting because I was able to use the popular press article and follow it back to the newspaper article and original blog post detailing the language technology and the press coverage revolving around this tool occurred the same day the blog post was published. It was like a game of “telephone” regarding the Amazon software.

The *C Net* article describes the software, Amazon Comprehend Medical, as a machine learning tool that can extrapolate key information from medical records so that key information could be recognized and allow for potentially less expensive, more efficient patient care. The article highlighted aspects of machine learning and gives an example of how neural networks can learn with high accuracy through text data. A key feature of the software is that it has better privacy than other software because data processed by the software will not be stored nor will be used in the training process. The *C Net* article highlighted the machine learning and applications of the software in their article.

¹ <https://www.cnet.com/news/amazon-plans-to-sell-software-that-reads-medical-records/>

² <https://www.wsj.com/articles/amazon-starts-selling-software-to-mine-patient-health-records-1543352136>

³ <https://aws.amazon.com/blogs/machine-learning/introducing-medical-language-processing-with-amazon-comprehend-medical/>

With *C Net*, the title “Amazon plans to sell software that reads medical records” is pretty accurate. It does seem like a slight exaggeration though. Because I am familiar with machine learning and natural language processing, I naturally assumed that there was text parsing that was learned in the software. However, if a lay person read the title, they would likely assume that there is reading and comprehension done in a similar way that humans read text, because that is what the title literally dictates. Either way, the title is mostly accurate to the content regarding the software. The title describes in layman term what the software is attempting to do and that is what the content is about.

The *WSJ* article is essentially the same information provided by *C Net*, except it is more detailed with more information from interviews from informaticians and those working on the Amazon Comprehend Medical project. The highlighted point in this article is that the software essentially “reads” medical records to organize the information and return it in a cleaner format. The *WSJ* also gives an example about how the Fred Hutchinson Cancer Research Center is using this tool to recruit cancer patients in clinical trials. This article was under the “Business”, “Health Care”, and “Health” sections of the paper so there was a balance between talking about the benefits within the healthcare system along with cost effect of the given software.

With the *WSJ*, the title “Big Tech Expands Footprint in Health” is broad, but it does lure interest because after reading the title, it made me curious about the “which big tech?” and “how is health care being expanded?” In a way, the *WSJ* probably was overtly vague to cue reader interest. If I only read the title without reading the content, I would most likely just think that tech companies had interest in utilizing healthcare data, but I would have no idea how or what they were doing.

The original Amazon blogpost was an announcement of the software. The highlight of the blog post were the use cases, explanations of privacy and HIPAA compliance, and how machine learning was maximized in structuring data. The blog also gave the example of how the Fred Hutchinson Center was

utilizing the software. Reading the blogpost, it came across more of an advertisement so show how Amazon Comprehend Medical could be used in the health decision making process.

There were differences amongst the three sources detailing the software; Amazon was trying to sell the product, the *WSJ* highlighted the business and health aspects, while *C Net* primarily highlighted the technology aspect. It was clear that the *WSJ* did more research within the business and cost aspect because the Amazon blogpost did not go over exact numbers. *C Net* used more flashy terms to highlight that machine learning was used in this task of parsing through text to acquire the most useful information. However, it took me a minute to figure out that Amazon Comprehend Medical does NOT necessarily use neural networks to train their models; *C Net* brought up neural networks in their articles and how they are good for within text related tasks. I think I was originally misled because with machine learning and NLP in mind, I was trying to figure out the tool that was used in the machine learning and because they gave examples regarding spam detection with neural networks, I made the jump that the Amazon software used it, too, even though that was not explicitly written.

Within all these sources, it was clear that there was no experimental work or analysis was included; that information is most likely proprietary which is why we do not know any of the performances for model evaluation of Amazon Comprehend Medical. Within the *C Net* and the *WSJ* articles, there was no way to tell what kind of evaluation was carried out. If this information was available, I would be interested in finding out what type of data is fed into the software and how it can learn without storing any information; I would want to know more specifics about this aspect of the software. As a reader, I was able to determine that the software can parse medical text, but I really do not know any of the details on how it is done or exactly how good the software is. I feel like there is no difference between whether a linguist or a layman read the articles. It is only a brief description of what the software is attempting to do, but we really do not have concrete, numerical proof! However, I am sure that a linguist or someone versed with machine learning would also be in the same boat as me and

be curious on what algorithm is used and the exact experimental work that was carried out to create the final product.

None of the articles nor blogpost highlighted the limitations of the software. They talked about the uses and applications, but none of the shortcomings. I would be interested to see how much the information is culled down to (whether it is key terms versus short descriptions) after text parsing. I find it weird that there is this software being advertised (whether commercially or just to be known to the general public), but no one really knows exactly how it is being done. Based on the ethics lecture in class, I am aware now about biases in linguistics and would want to know more about the software in terms of how it was developed and the exact workings of it, especially if I were the person who was responsible for whether Amazon Comprehend Medical was to be implemented at a hospital or healthcare setting. Aside from accuracy, I would be curious what types of diagnoses or key terms are missed within the software and how it can handle negatives (ie: patient does NOT have meningitis).

Either way, all three sources seem pretty superficial in describing the software. I am left understanding the purpose of the software without really understanding it or having concrete evidence on its performance, so I cannot even compare it to other software that attempt to carry out a similar task. All the articles and blogposts attempt to reel in readers and make the headlines interesting, but if I were to be frank, I would retitle it as “Amazon develops software that can parse and structure medical record data”. It is very boring, but transparent and without exaggeration compared to the articles. I am not even sure if I could do a better job compared across the blogpost and the two articles, but if I were to create a social media post regarding the software, I imagine I would say something as following (probably a Facebook post... my post is longer):

“Recently I learned, Amazon develops the software, Amazon Comprehend Medical, which utilizes machine learning and text parsing to highlight key components of medical records. This allows for structured summarization of medical records, which could have broader implications within the healthcare field. Useful applications include clinical decision making and clinical trial recruitment. Although it is proprietary, I would definitely be interested in knowing exactly how this works! Amazon just tells us that this software is good! Strange that companies sell you products and hope that you trust it to work! I wonder how many things in life we take for granted in a similar way, trusting that it works without understanding how.”